Carbon-Aware Green Traffic Monitoring using Motion-based Deep Learning Model Scheduling

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***Abstract*—The growing use of intelligent traffic monitoring systems has increased the energy requirements of ongoing vehicle detection using deep learning algorithms. Traditional systems carry out object detection all the time, which causes extra computations as well as energy use and carbon emissions if there is no traffic at all. This paper aims to introduce an approach to adaptively schedule vehicle detection based on lightweight motion detection, which triggers deep learning inference only when necessary. The proposed framework assesses five deep-learning models: YOLOv8n, YOLOv9m, YOLOv11n, YOLOv26m, RT-DETR-L, and RF-DRTR-M both under continuous and motion-triggered inference modes. Energy consumption, power consumption, CO_2 emissions, energy per object detected, electricity cost, and object detection performance are used as criteria to evaluate the models. The suggested scheduling strategy led to energy savings of 40.11%–80.52% and carbon footprint reduction of 40.15%–85.72% for all the investigated object-detection models. The obtained findings show that model scheduling on the basis of motion can be highly effective for achieving high levels of energy efficiency and sustainability of traffic management systems.**

***Index Terms*—Internet of Things, energy-efficient traffic monitoring system, motion-based scheduling**

I. Introduction

The rapid growth of urbanization and road transportation has created an increasing demand for intelligent and automated traffic-monitoring systems. Continuous monitoring of road environments is important for applications such as vehicle detection, traffic-flow analysis, road surveillance, congestion monitoring, and intelligent transportation systems (ITS). In recent years, deep learning-based object detection has become an important component of automated traffic-monitoring systems because it can identify and classify vehicles directly from video streams with high accuracy [1], [2]. Modern object detectors, including the YOLO family and transformer-based architectures, have further improved the capability of real-time visual analysis by providing a balance between detection accuracy and inference speed [3]. However, deploying these models for continuous video monitoring introduces a significant computational challenge because the detector is typically executed on every incoming frame, regardless of whether meaningful traffic activity is present.

Continuous deep-learning inference can result in substantial computational and energy consumption. This issue becomes particularly important in long-duration video-monitoring applications, where cameras may operate continuously for several hours or even days. Neural-network inference is computationally expensive when applied to high-volume video streams, and previous research has demonstrated that exploiting temporal redundancy in video can substantially reduce the amount of unnecessary neural-network computation [4]. In conventional frame-by-frame inference, however, the same computationally intensive detection process is repeatedly performed even when consecutive frames contain little or no meaningful change. Therefore, continuously executing a deep-learning detector during inactive or low-activity periods can lead to unnecessary resource utilization. The environmental impact of computationally intensive artificial intelligence has also become an important research concern. Previous studies have emphasized that the increasing computational requirements of deep learning can lead to considerable energy consumption and associated carbon emissions [5]. Similarly, the Green AI perspective argues that computational efficiency should be

considered alongside predictive performance when evaluating AI systems, rather than focusing exclusively on accuracy [6]. These considerations are particularly relevant for continuously operating computer-vision applications because the environmental cost is not limited to model training; repeated inference over large numbers of video frames can also contribute to the overall energy footprint of an AI-enabled monitoring system.

Traffic-monitoring systems provide a particularly suitable environment for reducing unnecessary inference because road scenes generally contain periods of both activity and inactivity. During periods of active traffic, continuous object detection is necessary to identify vehicles and analyze the scene. In contrast, during periods with little or no movement, executing a computationally expensive object detector on every frame may provide limited additional information. This temporal variation creates an opportunity to introduce an adaptive inference mechanism in which a lightweight computational process first determines whether meaningful activity is present and activates the deep-learning detector only when required. Similar principles have been investigated in video-analytics systems, where temporal differences and specialized lightweight models are used to avoid unnecessary execution of expensive neural networks [4]. However, the application of motion-based model scheduling specifically for reducing the energy and carbon footprint of continuous traffic monitoring remains an important area for investigation.

Recent advances in real-time object detection have produced a variety of models with different computational and accuracy characteristics. YOLO-based detectors are widely used in real-time applications because of their efficient inference capabilities, while transformer-based approaches have introduced alternative detection architectures. RT-DETR, for example, is an end-to-end real-time transformer-based object detector designed to achieve a favorable balance between detection accuracy and inference speed [3]. Although improvements in detector architecture can reduce the computational cost of individual inference operations, continuously executing even an efficient detector still requires computation for every processed frame. Therefore, improving the detector alone does not completely address the problem of unnecessary inference during periods of low scene activity.

This observation motivates the development of a motion-based deep-learning model scheduling strategy for energy- and carbon-conscious traffic monitoring. Instead of applying the selected object detector to every video frame, the proposed framework introduces a lightweight motion-detection stage before deep-learning inference. The motion-detection stage continuously analyzes the scene and determines whether meaningful movement is present. When sufficient motion is detected, the deep-learning object detector is activated to perform vehicle detection. When meaningful motion is absent, deep-learning inference is temporarily suspended, allowing the system to operate in a lower-computation monitoring state. In this way, the proposed approach aims to preserve the detection capability required during active traffic periods while avoiding unnecessary deep-learning computation during inactive periods.

The proposed framework evaluates this scheduling strategy using five deep-learning object-detection models: YOLOv8n, YOLOv9c, YOLOv11n, YOLOv26m, RT-DETR-L, and RF-DETR-M. The models are evaluated under both continuous-inference and motion-based scheduling configurations. In addition to conventional object-detection performance, the study evaluates energy consumption, average power consumption, CO_2 emissions, electricity cost, and other efficiency-related measures to quantify the environmental and computational benefits of adaptive model execution. This evaluation provides a broader perspective on the trade-off between detection performance and computational sustainability.

Although previous studies have investigated energy-efficient artificial intelligence, video-analytics optimization, and intelligent traffic monitoring [1], [2], [4], [6], comparatively less attention has been given to explicitly controlling the execution of the underlying vehicle detector according to scene activity while simultaneously evaluating the resulting energy and carbon implications. Rather than modifying the internal architecture of an object detector, this work investigates an external scheduling mechanism that can be applied to different detection models. Such a strategy can potentially complement advances in model architecture by reducing the number of unnecessary inference operations performed during continuous monitoring.

The main contributions of this work are as follows:

- **Motion-based deep-learning model scheduling:** A lightweight motion-based scheduling framework is proposed to dynamically control the execution of a deep-learning vehicle detector, activating inference during meaningful traffic activity and suspending it during inactive periods.
- **Multi-model evaluation:** The proposed scheduling strategy is systematically evaluated using five real-time object-detection models, namely YOLOv8n, YOLOv9c, YOLOv11n, YOLOv26n, and RT-DETR-L, providing a comparative analysis across different detector architectures and computational characteristics.
- **Energy and carbon-efficiency analysis:** The study evaluates the impact of motion-based scheduling on energy consumption, average power consumption, CO_2 emissions, electricity cost, and related efficiency measures to quantify the environmental benefits of reducing unnecessary deep-learning inference.
- **Detection performance and efficiency trade-off:** The work investigates the relationship between object-detection performance and computational savings under continuous and scheduled inference configurations, demonstrating how model execution can be reduced while retaining the ability to detect vehicles during active traffic periods.
- **Practical framework for sustainable traffic monitoring:** The proposed approach provides a model-independent scheduling strategy that can complement existing real-time object detectors and offers a practical

direction toward more energy-efficient and environmentally sustainable continuous traffic-monitoring systems.

The experimental results demonstrate that motion-based model scheduling can substantially reduce the computational and environmental overhead associated with continuous traffic monitoring. By avoiding unnecessary execution of the deep-learning detector during periods of low or absent traffic activity, the proposed approach reduces energy consumption and CO_2 emissions across the evaluated models. The results further demonstrate that the magnitude of the savings depends on the characteristics of the selected detector, highlighting the importance of considering both detection performance and computational efficiency when designing sustainable AI-based monitoring systems.

The remainder of the paper is organized as follows. Section II presents the proposed related work. Section III describes the methodology of the research. Section IV contains the experimental results and discussion. Section V presents the future work and finally, Section VI contains the conclusion of the paper.

II. Related Work

In this section, we survey the previous literature related to energy- and carbon-aware traffic monitoring. Research on energy- and carbon-aware traffic monitoring draws on several overlapping strands: (i) energy-efficient object detection on resource-constrained edge devices, (ii) adaptive and carbon-aware traffic management frameworks, and (iii) inference-scheduling or frame-skipping strategies that avoid unnecessary computation in video-based vehicle detection. This section involves three parts. The first discusses energy-efficient object detection approaches for resource-constrained edge and IoT devices. The second concerns adaptive and carbon-aware traffic management frameworks for sustainable intelligent transportation systems. The third focuses on inference-scheduling and frame-skipping strategies that reduce unnecessary computation in video-based vehicle detection.

A. Energy-Efficient Object Detection on Edge and IoT Devices

The rapid growth in the computational cost of deep learning and the associated energy use and carbon footprint has led to calls to treat efficiency as a first-class evaluation criterion alongside accuracy [5], [6]. In response, a number of studies have looked into ways to lower the energy consumption of deep learning-based object detectors when they are used with limited edge hardware. Evaluated on an AMD Alveo U50 FPGA, a Jetson Orin Nano, and a Hailo-8 accelerator, the work in [7] suggested a hybrid fast-moving object identification pipeline that combines lightweight frame differencing with an AI classifier (MobileNet, ResNet50, Inception-v4, or ViT-Base). The hybrid solution produced a 28.3% average accuracy improvement, an 3.6 efficiency gain, and a 39.3% latency reduction when compared to an end-to-end YOLOX baseline. MobileNet consistently produced the best accuracy-latency-energy trade-off [7]. This demonstrates that an efficient method for energy-constrained inference is to separate a lightweight triggering mechanism from a heavier classification or detection stage. This idea is directly related to the motion-gated design used in this work. The same concept underpins cascade-style video analytics systems such as NoScope [4], which reports speed-ups of up to three orders of magnitude on fixed-camera video while maintaining accuracy within 1–5% of the reference model by positioning low-cost difference detectors and specialized models in front of an expensive reference network.

The EAAEIN framework [8] addresses the more general issue that the majority of edge-AI energy-efficiency promises are theoretical (FLOPs or parameter counts) rather than quantified. The authors of [8] contend that such reductions do not accurately measure the energy actually used during live inference, despite the fact that FLOPs have been suggested as a hardware-agnostic proxy for efficiency [6]. EAAEIN achieved 96.68% accuracy at only 0.43 J per inference, a 62.11% energy savings over Tiny-YOLO across Raspberry Pi, Jetson Nano, and TinyML platforms [8] by combining dynamic neural pathway pruning, quantization-aware inference, workload-sensitive voltage–frequency scaling, and real-time energy feedback control. Even though EAAEIN works at the architecture/runtime level rather than the scene level, it supports the project's larger goal, which is that energy savings should be the standard for green AI systems.

The benchmarking review of YOLOv8 and RT-DETR [3] on edge devices [9], which is most directly related to the current study, assessed both large detectors on a Raspberry Pi 5 and an NVIDIA Jetson Orin NX using the COCO val2017 dataset across multiple runtimes (PyTorch, TensorRT, TensorFlow Lite, HailoRT, and NCNN). The best configuration (YOLOv8l on Hailo-8) only achieved 1.73 FPS/W, and none of the evaluated setups met the 25 FPS threshold needed for a complete video pipeline [9]. The study discovered that big detectors struggle to achieve real-time throughput on edge hardware. The accuracy-speed trade-off between small and big detectors on edge hardware is demonstrated by the lightweight YOLOv7-tiny and YOLOv8-nano detectors deployed in [1], which achieved 23-35 FPS at 82-83.7% mAP on Jetson Nano or Xavier NX. Crucially, this work measures the energy cost of RT-DETR on edge devices but does not suggest a way to prevent RT-DETR from operating when there is no motion or vehicle in the scene. This is the exact gap that the current study fills: instead of optimizing the runtime or quantization of RT-DETR, the suggested system gates *when* RT-DETR runs at all using a lightweight motion detector (MOG2).

B. Adaptive and Carbon-Aware Traffic Management Systems

Instead of focusing on detector-level efficiency, a second line of inquiry aims to reduce energy and carbon at the traffic system level. In addition to a Passenger-Car-Equivalent (PCE) weighted density estimator and a rule-based control algorithm with starvation prevention, an edge-deployed real-time Adaptive Traffic Light Control (ATLC) system [1] benchmarks YOLOv7-tiny, YOLOv8-nano, YOLOv8-small, and YOLO-NAS-small for vehicle detection. The system achieved up to

90% mAP at 74 FPS and decreased junction congestion by up to 33% and waiting time by 16.6–23.4% in comparison to fixed-time signals [1]. It was deployed on Jetson Nano/Xavier NX and an Arduino Portenta H7 and evaluated on a custom Karachi traffic dataset. This system's focus is on signal-timing optimization rather than lowering the energy cost of the detection model itself, but unlike a lot of previous ITS research, it was proven on actual hardware rather than only in simulation. Both the delay reductions reported for learning-based signal control in [11] and the multi-criteria evaluation in [10], where Adaptive Signal Control rated top among possible interventions, support the value of adaptive signalling.

The AGMO-TSC framework [11] reduces average intersection delay by 31.4% and transport-sector CO_2 emissions by 27.8% at the system level by integrating four AI components into a single pipeline: STGAT for traffic prediction, MORL-SC for signal control, GNN-EVR for EV routing, and RTCA for real-time carbon accounting. In a similar vein, a 6G-enabled IoT-based ITS [12] reports a 29% increase in traffic flow efficiency and 32-47% reductions in CO_2, NO_x, and $PM_{2.5}$ emissions along with a 94.1% congestion-prediction accuracy [12] by combining THz-band 6G communication, edge computing, and a Deep-Q-Network-based carbon-aware routing algorithm. In the policy domain, a design-science-research-based framework for Indian smart cities [10] ranked five potential interventions using Monte Carlo robustness analysis and multi-criteria decision-making (entropy weighting, AHP, TOPSIS), concluding that adaptive signal control was the most reliable option for simultaneously optimizing emissions and road safety [10].

Although there is a definite trend toward carbon-aware traffic management in these four studies [1], [10]–[12], each one approaches the vehicle-detection or sensing component as a fixed, constantly operating module rather than as a target for energy optimisation in and of itself. None of them look at the particular technique suggested in this work—scheduling or deactivating the underlying deep-learning detector itself, depending on scene activity.

C. Inference Scheduling and Frame-Skipping for Vehicle Detection

Since consecutive frames from a stationary camera frequently change very little, video-based detection is intrinsically redundant. Frame differencing has been employed as such a trigger in [7] and [4]. Traditional background-subtraction techniques based on Gaussian mixture models [13]–[15] offer a very affordable, non-deep-learning cue for whether anything in the scene has changed. A hybrid conditional frame-skipping strategy for YOLOv8n-based vehicle detection [2], which combines motion-aware conditional inference, periodic detector refresh (to prevent accumulated error), and online threshold calibration, is the work that most closely aligns methodologically with the current study. With a tiny 0.0008 decline in F1 score [2], the optimal configuration increased throughput from 58.7 to 88.7 FPS and reduced mean GPU energy consumption by 34.3% when tested on 40 movies from the UA-DETRAC dataset. This demonstrates that motion-conditioned inference may significantly lower energy usage without significantly compromising accuracy in video-based vehicle detection.

However, [2] deviates from the current work in three significant ways: (1) it *skips frames* within a continuously active detector instead of completely deactivating the detector during vehicle-free periods; (2) its energy evaluation is restricted to GPU platforms, leaving open the question of edge-device applicability; and (3) it does not evaluate carbon-emission reduction as a sustainability metric, focusing only on GPU energy in joules. Thereby completely gating RT-DETR activation on a dedicated, inexpensive MOG2 motion detector [14], [15] instead of periodically skipping frames within an always-on detector—and by explicitly framing the resulting energy savings in terms of carbon-emission reduction, the current study expands on this line of work.

D. Synthesis and Research Gap

Two families of contributions emerge from the eight reviewed studies: (i) energy-efficient model or runtime design for edge inference [7]–[9] and (ii) system-level, carbon-aware traffic management [1], [10]–[12], with a smaller but closely related body of work on inference scheduling for video-based detection [2]. Interestingly, [2] suggests a scheduling-like mechanism (frame skipping) for YOLOv8n instead of RT-DETR and without a carbon-emission framing, whereas [9] explicitly benchmarks RT-DETR's energy cost on edge devices without proposing a scheduling mechanism to avoid unnecessary execution.

In a single traffic-monitoring pipeline, no evaluated work integrates (a) a high-accuracy transformer-based detector like RT-DETR, (b) a lightweight, classical (non-deep-learning) motion detector like MOG2 functioning as an activation gate, and (c) an explicit carbon-emission-reduction aim. The suggested system fills this gap by benchmarking RT-DETR against DINO, RF-DETR, YOLOv9, YOLOv11, and YOLO26 in order to support the selection of the detection backbone prior to implementing the suggested motion-based scheduling method.

Our proposed framework addresses critical limitations observed in existing energy-aware detection and traffic-management studies. Most edge-oriented works either pair a low-cost trigger with a lightweight classifier instead of a high-accuracy detector, or focus on compressing or optimizing the detection model or its runtime. Although they regard the car detector as an always-on component, system-level traffic-management frameworks improve emissions and congestion. The closest method now in use, conditional frame skipping, does not include carbon emissions as a design goal and merely saves computation by skipping frames inside an active detector. marks, A carbon-aware traffic-monitoring pipeline that only activates a high-accuracy transformer-based detector when the scenario calls for it is still lacking. Our work fills this gap by placing a lightweight, CPU-friendly MOG2 motion detector in front of RT-DETR so that the detector is

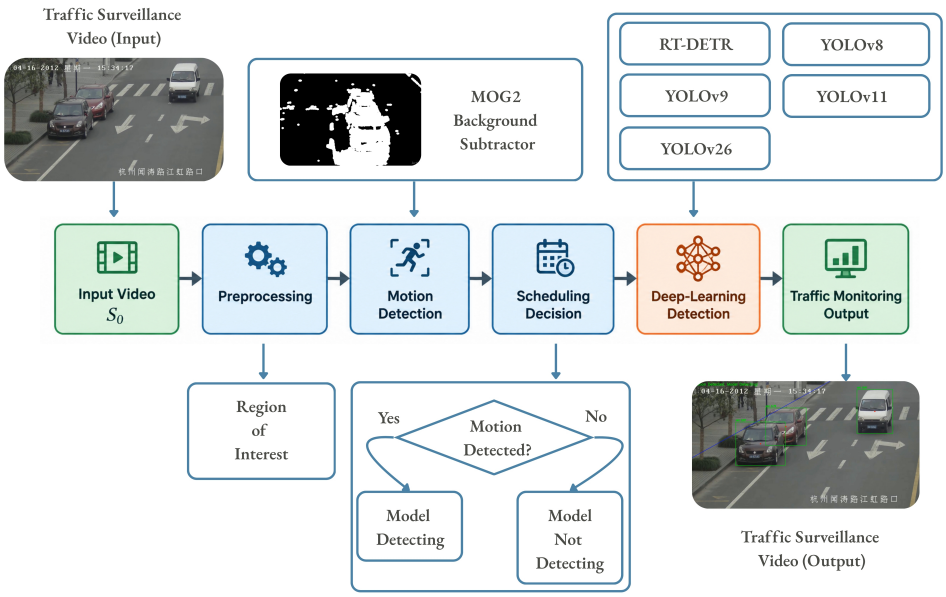


Fig. 1: Overview of the proposed traffic monitoring framework.

activated only when motion is present and is fully deactivated during vehicle-free periods. This differs from frame skipping by completely turning off the detector instead of sampling less often, and it differs from system-level frameworks by focusing on the detection model's energy cost. A motion-gated scheduler eliminates this waste while maintaining the precision that initially drives the usage of a transformer-based detector, which is why such a design is necessary. Continuous inference loses compute, energy, and carbon anytime the road is empty.

III. Methodology

The proposed system for carbon-aware traffic monitoring uses a motion-based deep learning model scheduling scheme to avoid unnecessary computation and energy consumption. Rather than running the object detection model on every frame of the video sequence, a lightweight motion detection stage first determines whether meaningful activity is present in the monitored scene. The deep learning detector is triggered when sufficient motion is detected. When meaningful motion is absent, deep learning inference is paused. This design follows the general principle of exploiting temporal redundancy in video to avoid repeatedly executing computationally expensive neural-network inference [4].

A. System Overview

The proposed system combines motion detection and deep learning object detection to reduce unnecessary computation in continuous traffic monitoring. As shown in Fig. 1, the system takes a traffic surveillance video as input and first performs preprocessing to define the region of interest (ROI). The preprocessed frames are then analyzed using the MOG2 background-subtraction algorithm. MOG2 is an adaptive Gaussian mixture model-based background-subtraction method designed to identify foreground changes in video scenes [14]. The detected motion information is subsequently provided to the scheduling component, which determines whether the deep learning detector should be activated.

If sufficient motion is detected within the ROI, the selected object detection model RT-DETR, YOLOv8, YOLOv9, YOLOv11, or YOLOv26m is activated to perform vehicle detection. When meaningful motion is not detected, the deep learning inference stage remains inactive, thereby avoiding unnecessary detector execution. This motion-triggered execution strategy is related to prior video-analytics approaches that exploit temporal changes to reduce the computational cost of frame-by-frame deep learning inference [2], [4].

B. Video Preprocessing and Region of Interest

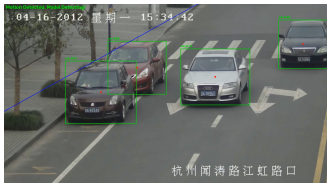
Surveillance video frames form the input to the system. For each input frame, a region of interest is defined to cover the relevant road area. Unwanted regions, such as buildings, sky, and roadsides, are excluded from the ROI so that motion analysis is concentrated on the monitored traffic region. Restricting background-subtraction analysis to the relevant scene area helps reduce the influence of irrelevant background changes and provides a more targeted motion signal for the scheduling stage.

C. Motion-Based Model Scheduling

The scheduling component analyzes successive ROI frames using the MOG2 background-subtraction method. MOG2 models the background using an adaptive mixture of Gaussian distributions and identifies pixels that differ sufficiently from the estimated background [14]. The resulting foreground mask is used to estimate the amount of motion present within the monitored ROI.

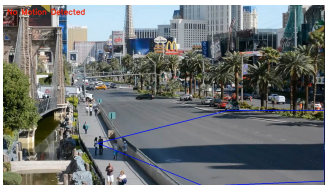


(a) No motion detected

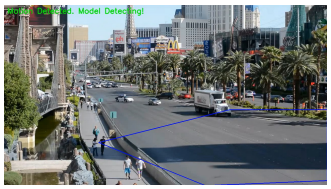


(b) Motion detected

Fig. 2: Comparison of traffic surveillance under different motion conditions.(Scenario 1)



(a) No motion detected



(b) Motion detected

Fig. 3: Comparison of traffic surveillance under different motion conditions.(Scenario 2)

A predefined motion threshold is then applied to the detected foreground activity. If the measured motion exceeds the threshold, the scheduler generates an activation signal and enables the deep learning detector. If the measured motion remains below the threshold, the detector is not executed for that frame. This conditional execution mechanism is motivated by the observation that consecutive frames in fixed-camera video often contain substantial temporal redundancy, making continuous execution of an expensive detector unnecessary during periods of little scene change [4].

Importantly, the motion detector acts only as a lightweight activation mechanism and does not replace the deep learning object detector. The purpose of the scheduler is to determine *when* the detector should operate, while the selected deep learning model continues to perform the actual vehicle detection whenever the scheduler activates it.

D. Deep Learning-Based Vehicle Detection

When sufficient motion is detected in the monitored region, the selected deep learning object detector processes the corresponding video frame. The detector identifies vehicles and produces their corresponding bounding boxes and confidence scores. The evaluated detectors include RT-DETR, YOLOv8, YOLOv9, YOLOv11, and YOLOv26.

When the motion-based scheduler determines that meaningful activity is absent, the detector is temporarily deactivated and the system remains in the low-computation monitoring state. Therefore, the proposed approach does not modify the architecture or parameters of the individual object detectors; instead, it operates as an external scheduling layer that controls detector invocation. Similar detector-independent scheduling principles have been investigated in recent energy-efficient video detection research, where motion-aware conditional inference is used to reduce unnecessary detector execution [2].

E. Energy and Carbon Emission Measurement

The effectiveness of the proposed scheduling strategy is evaluated by comparing continuous detector inference with motion-based scheduled inference. The two configurations are compared in terms of energy consumption, average power consumption, CO_2 emissions, inference performance, and vehicle detection accuracy.

For continuous inference, the selected deep learning detector is executed throughout the monitoring period. For motion-based inference, the detector is activated only when the motion scheduler detects sufficient activity in the ROI. The difference between the two configurations is then used to quantify the energy and carbon benefits obtained by reducing unnecessary deep learning inference. This evaluation follows the broader motivation of energy-efficient video analytics, where reducing redundant neural-network execution can substantially decrease computational workload [2], [4].

IV. Results

A. Experimental Evaluation

The performance of the proposed traffic monitoring system using the carbon-aware technique has been investigated to examine the possibility of employing motion-based model scheduling to lower computational energy and the carbon emission related to this process while keeping the ability to detect objects in traffic continuously. The fundamental idea behind this approach is that the object-detection model will be active only in case of a moving car in the monitored road scene. Otherwise, the model will not be executed because the computationally costly process of detection will not take place. Five object-detection models (RT-DETR, YOLOv8, YOLOv9, YOLOv11, and YOLOv26) have been investigated. It could be seen that scheduling works effectively in an environmental and computational sense.

B. Motion-Based Model Scheduling Performance

Scheduling led to a reduction in energy consumption for all analyzed models. In the absence of any scheduling, RT-DETR would have used 29.335 Wh, but with scheduling, this dropped to 5.713 Wh. Likewise, energy usage by YOLOv8 fell from 5.12 Wh to 1.54 Wh, YOLOv9 fell from 6.056 Wh to 1.345 Wh, YOLOv11 from 3.219 Wh to 1.928 Wh, and YOLOv26 fell from 2.738 Wh to 0.856 Wh.This clearly demonstrates the effectiveness of the approach in preventing any need for inference at times when no moving vehicles are around. It should be noted that this approach does not involve any changes to the trained detection model itself or its operation.

C. Detection Performance Comparison

From the results on detection performance on the first scenario, we find that the suggested concept of scheduling can be implemented without any compromise on the core detection capabilities. RT-DETR had the highest value for mAP@50 (0.99), along with recall of 1.00, while YOLOv11 had mAP@50 of 0.97, F1-Score of 0.98, and recall of

TABLE I: Performance comparison of the evaluated object detection models. (Surveillance scenario 1)

Model	mAP@50	F1-Score	FPS	Recall	Parameters (M)
RT-DETR	0.99	0.99	43	1.00	32.0
YOLOv8	0.94	0.94	57	0.93	3.15
YOLOv9	0.96	0.96	32	0.97	25.5
YOLOv11	0.97	0.98	30	0.99	2.62
YOLOv26	0.98	0.96	30	0.95	22.4
RF-DETR	0.98	0.97	38	0.98	33.7

TABLE II: Performance comparison of the evaluated object detection models. (Surveillance scenario 2)

Model	mAP@50	F1-Score	FPS	Recall	Parameters (M)
RT-DETR	0.99	0.98	37	0.99	32.0
YOLOv8	0.93	0.91	54	0.94	3.15
YOLOv9	0.96	0.93	38	0.94	25.5
YOLOv11	0.97	0.95	36	0.98	2.62
YOLOv26	0.99	0.96	32	0.97	22.4
RF-DETR	0.97	0.98	39	0.97	33.7

0.99. The YOLOv8 had mAP@50 and recall of 0.93 each, whereas YOLOv9 had mAP@50 and recall of 0.96 and 0.97, respectively.Hence, RT-DETR has shown to have the best detection performance, while YOLOv11 has emerged as the most appealing option for its detection accuracy and model simplicity.

Comparison Table II for the second scenario, summarizes the performance of the detection algorithms by considering accuracy, efficiency, and model complexity. RT-DETR and YOLOv26 algorithms have the highest mAP@50 score of 0.99 and RT-DETR also has the highest recall score of 0.99. YOLOv8 has the highest efficiency score of 54 FPS with only 3.15M parameters. YOLOv11 has an efficient combination of accuracy and compactness, and RF-DETR has a high F1-score of 0.98.

D. Energy Consumption Analysis

There was also a reduction in carbon emissions in the first scenario. The RT-DETR had carbon emissions of 18.844 g without scheduling and only 2.690 g with scheduling, resulting in an 85.72% reduction. On the other hand, YOLOv9 had an 81.31% reduction, YOLOv26 had an 72.41% reduction, YOLOv8 had a 70.22% reduction, and YOLOv11 had a 40.15% reduction.

Overall, the reduction in carbon emissions by all five models was about 69.96%. Thus, these results prove that reducing unnecessary computations could result in environmental gains, and hence event-based model execution is appropriate for green traffic monitoring applications.

Model scheduling in the second scenario leads to significant reductions in energy consumption for all considered models. RT-DETR attains the highest percentage of energy savings of 79.85%, cutting down energy consumption from 18.42 Wh to 3.711 Wh. YOLOv9 obtains a 77.70% reduction in energy savings, followed by YOLOv8 with a 65.53% reduction in energy savings. The percentages of energy savings for RF-DETR and YOLOv26 are 61.85% and 57.00%, respectively.

E. Carbon-Emission Analysis

Due to the direct relationship between electricity usage with the operational carbon footprint of the monitoring system, any changes in the electricity usage will be also seen in the emitted CO_2 amount. The test results prove that there is a significant decrease in the case of the YOLOv11n model: the amount of emitted CO_2 was reduced from 0.919 g in the unscheduled experiment to 0.550 g in the scheduled experiment, which corresponds to approximately a 40.1% decrease. In the case of RT-DETR-L, there is also a considerable difference between the two reported models: the scheduled experiment shows 2.690 g of CO_2 emissions while in the unscheduled one the amount is 18.844 g. Thus, the carbon results also demonstrate a similar pattern to the energy results. Any decrease in electricity usage due to the chosen scheduling configuration causes a decrease in the emissions. This helps to achieve the carbon-aware goal of the proposed framework. But the results produced by YOLOv8 and YOLOv9 have shown that carbon savings are not guaranteed simply based on the inclusion of a scheduler. YOLOv8 was found to consume more energy and

TABLE III: Energy Consumption and Carbon Emission Comparison With and Without Model Scheduling (Surveillance scenario 1)

Model	Configuration	Energy (Wh)	Avg. Power (W)	CO_2 Emission (g)	Energy Saved (%)	Carbon Emission Decreased (%)
RT-DETR	Without Scheduling	29.335	13.71	18.844	80.52%	85.72%
	With Scheduling	5.713	71.00	2.690		
YOLOv8	Without Scheduling	5.12	44.07	1.78	69.92%	70.22%
	With Scheduling	1.54	46.71	0.53		
YOLOv9	Without Scheduling	6.056	51.07	0.840	77.70%	81.31%
	With Scheduling	1.345	44.56	0.157		
YOLOv11	Without Scheduling	3.219	13.75	0.919	40.11%	40.15%
	With Scheduling	1.928	13.85	0.550		
YOLOv26	Without Scheduling	2.738	13.97	0.380	64.74%	72.41%
	With Scheduling	0.856	11.67	0.104		
RF-DETR	Without Scheduling	25.31	12.79	9.92	70.13%	80.95%
	With Scheduling	7.56	15.81	1.89		

TABLE IV: Energy Consumption and Carbon Emission Comparison With and Without Model Scheduling (Surveillance scenario 2)

Model	Configuration	Energy (Wh)	Avg. Power (W)	CO_2 Emission (g)	Energy Saved (%)	Carbon Emission Decreased (%)
RT-DETR	Without Scheduling	18.42	9.73	14.79	79.85%	81.54%
	With Scheduling	3.711	11.67	2.73		
YOLOv8	Without Scheduling	3.79	39.46	1.78	65.53%	68.34%
	With Scheduling	1.32	43.74	1.22		
YOLOv9	Without Scheduling	4.78	48.69	0.793	77.70%	84.89%
	With Scheduling	1.345	47.97	0.122		
YOLOv11	Without Scheduling	1.832	13.75	0.675	54.33%	62.07%
	With Scheduling	0.854	13.85	0.256		
YOLOv26	Without Scheduling	1.563	13.97	0.234	57.00%	64.95%
	With Scheduling	0.672	11.67	0.082		
RF-DETR	Without Scheduling	16.57	12.79	7.98	61.85%	62.78%
	With Scheduling	6.32	15.81	2.97		

emit more CO_2 while using scheduling than without, whereas YOLOv9 did not change in either respect at all. Thus, the carbon savings are contingent upon the reduction of expensive computations.

During the second scenario, the scheduling technique not only increases the efficiency of the models but also reduces their carbon footprint significantly. For instance, YOLOv9 shows the highest reduction rate of 84.89%, which reduces their carbon footprints from 0.793g to 0.122g. The RT-DETR model has a reduction rate of 81.54%, whereas YOLOv8 has a reduction rate of 68.34%. YOLOv26, RF-DETR, and YOLOv11 have reduction rates of 64.95%, 62.78%, and 62.07%, respectively.

F. Comparison Across Detection Models

RT-DETR had the highest detection accuracy but had the largest computation footprint, which was 32.0 M. YOLOv8 was capable of delivering 57 FPS with a total of 3.15 M, hence making it one of the most efficient models for real-time application. YOLOv9 had 25.5 M and 32 FPS, whereas YOLOv11 had 30 FPS with 2.62 M parameters. The YOLOv26 model had the least model size of 2.41 M with 30 FPS.

Hence, the choice of the model should be based on the detection quality and sustainable computation rather than just accuracy.

G. Energy–Accuracy Trade-off

The results indicate the existence of a vital trade-off between energy efficiency and accuracy. RT-DETR has excellent object detection performance but demands a lot of energy. On the other hand, the YOLOv11 has excellent object detection accuracy and uses only 2.62 M parameters, hence being much more efficient. The YOLOv8 is favorable because it has a record-breaking speed of 57 FPS and saves 69.92% energy.

The scheduling technique helps to improve the trade-off since it lowers the time taken by the model in operation without lowering the detection capacity of the model.

V. Discussion

The results of the experiments clearly show that scheduling taking into account motion and carbon emission is a powerful tool for increasing sustainability of intelligent traffic-monitoring systems. Instead of constantly performing the computation of heavy object detection models, the proposed solution activates the model in case the motion matching the movement of a vehicle is detected. Otherwise, the model will not be activated, which means no additional computations and energy usage as well as less carbon emissions. This observation is consistent with previous video analytics research showing that exploiting temporal redundancy can reduce the need for continuous execution of computationally expensive neural networks [4]. Recent work on energy-efficient vehicle detection has similarly demonstrated the potential of conditional inference and frame skipping to reduce computational and energy requirements [2].

Furthermore, the experiment results prove that the benefits of the scheduling depend on the architecture of object detection models used and vary in accordance with the amount of

computations required by the models. RT-DETR showed the best detection results; therefore, it is good to use in cases where the accuracy of the detection is the main priority. But the high number of parameters in the model requires special attention to energy optimization due to the number of computations. YOLOv8 and YOLOv11 offer a better trade-off between detection, speed of processing and computation power.

The most important part about this is that the suggested approach does not need any changes to the detection model used and thus can be used along with various models and their detection capabilities. Overall, it can be concluded that the intelligent management of the model execution process ensures proper trade-off between performance, efficiency, and sustainability of the process. Such detector-independent scheduling is consistent with the broader principle of reducing redundant neural-network execution in video analytics systems [4]. Overall, intelligent management of detector execution provides a practical trade-off between detection performance, computational efficiency, and environmental sustainability.

As can be seen from the experimental results, motion-based scheduling of the models used helps significantly increase the environmental efficiency of the traffic surveillance system. On average, in all five models tested, the scheduling provided energy savings of 66.60% and carbon emission savings of 69.96%. In terms of energy, the maximum savings were obtained for RT-DETR and amounted to 80.52%, while in terms of carbon emission, the maximum savings were again for RT-DETR and amounted to 85.72%. Importantly, during the experiment, the detection models maintained their detection capacity. Thus, it can be stated that the approach of activating object detection based on the movement of vehicles is a successful one. These results indicate that activating object detection conditionally based on scene motion can be an effective strategy for reducing the computational and environmental costs of continuous traffic surveillance, particularly in periods with limited scene activity [2].

VI. Limitation and Future Work

However, despite the effectiveness of the suggested carbon-aware scheduling strategy for energy reduction during computation and carbon dioxide emission, there are some disadvantages associated with the current model. The existing model scheduling depends on motion detection in order to initiate deep learning processing; therefore, there might be some cases where this detection will fail due to different issues, including lack of light, vehicle occlusion, or low motion. Besides, there is no real-time electricity carbon intensity applied in the current model, and its assessment is conducted only using the considered traffic scenarios and detection models.

An adaptive inference scheduler that changes the activation of the deep learning models depending on the traffic, motion, and confidence of detections. The suggested system can also be improved by using the carbon intensity of electricity in real time so that the inference will take place not only on the basis of computation but also on the environment. Future studies will explore the use of lightweight scheduling algorithms and several detection models for a better efficiency-to-energy ratio. Besides, further tests will take place in different traffic environments.

VII. Conclusion

In this paper, a carbon-aware adaptive vehicle detection model has been introduced, which aims to lower unnecessary deep-learning inference processes in the case of continuous traffic monitoring using inference scheduling according to the motion detection process. For performance evaluation, the proposed system was tested using five different models, including YOLOv8n, YOLOv9c, YOLOv11n, YOLOv26n, and RT-DETR-L in both continuous and adaptive modes. The obtained results indicate that the scheduling process of deep-learning inference based on the detection of traffic activities is capable of decreasing computational energy consumption, carbon emissions, and energy consumption per detection process. In particular, using YOLOv11n, an average reduction in energy consumption of about 40.1%, from 3.219 Wh to 1.928 Wh, and CO_2 emissions from 0.919 g to 0.550 g have been reported. It is important to note that the efficiency of adaptive scheduling depends on the model used and other factors.

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